Evidence-Based Study Methods for Adaptive Study Sessions

RAG-ready knowledge document for the EstudaUnB study agent
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Purpose. This document defines a small, auditable catalog of study methods, their evidence base, operational protocols, limits, and the data that an adaptive agent may collect after each study session. It is not a clinical or diagnostic instrument.

1. Design principles

Separate learning method from time-management format. Retrieval practice, spacing, interleaving, worked examples, and self-explanation are learning strategies. Pomodoro is primarily a time-structuring technique; it does not determine what cognitive activity happens inside the interval.

Do not optimize only for user preference. A method may feel easy or pleasant while producing weak retention. The agent should combine subjective ratings with objective indicators such as task completion and delayed self-test performance.

Context matters. The same method can perform differently for factual recall, conceptual understanding, programming, mathematics, writing, or project work.

Keep claims calibrated. The agent may recommend, compare, and explain. It must not claim that a method is universally best or infer a stable learning style.

Evidence labels used in this document

Label	Meaning
Strong	Supported by converging reviews, meta-analyses, or repeated experimental findings.
Moderate	Supported by credible studies, but effects depend strongly on task, implementation, or population.
Limited	Direct empirical evidence is sparse; use as an optional workflow aid, not as a proven learning mechanism.

2. Method catalog

Retrieval practice (practice testing / active recall)

Evidence: Strong

Purpose: Strengthen long-term access to previously studied information by attempting to retrieve it without looking at the answer.

Operational protocol

1. Define a narrow learning objective.
2. Close notes or hide the solution.
3. Answer questions, solve problems, explain from memory, or write a free-recall summary.
4. Check the answer and record errors.
5. Correct errors with feedback, then retry later.

Good fit: Facts, concepts, definitions, procedures, problem-solving steps, and exam preparation.

Limits and failure modes: Retrieval without feedback can reinforce errors. Very difficult prompts may require hints or worked examples before independent retrieval.

Agent rule: Prefer when the learner has already had initial exposure to the content. Track correctness, confidence, and delayed recall.

Distributed practice (spacing)

Evidence: Strong

Purpose: Distribute study and retrieval across multiple occasions instead of concentrating all practice in one block.

Operational protocol

1. Split a topic across several days or weeks.
2. Revisit material after some forgetting has occurred.
3. Use retrieval during each revisit rather than passive rereading only.
4. Increase intervals when performance is stable; shorten them after repeated failure.

Good fit: Durable retention, cumulative courses, vocabulary, formulas, concepts, and recurring exam preparation.

Limits and failure modes: There is no single universal interval. Scheduling depends on the retention horizon, difficulty, and prior performance.

Agent rule: Recommend future reviews before the assessment date. Never schedule a review after the relevant exam when the goal is exam preparation.

Interleaved practice

Evidence: Moderate

Purpose: Mix related categories or problem types so the learner must identify the appropriate strategy instead of repeating one procedure mechanically.

Operational protocol

1. Select two to four related problem types.
2. Alternate them in a mixed set.
3. Do not label the solution method in advance.
4. After each item, explain why that method applies.

Good fit: Mathematics, programming patterns, classification, diagnosis, and tasks requiring discrimination between similar cases.

Limits and failure modes: May initially feel harder and reduce immediate fluency. It is inappropriate when the learner lacks a basic worked example for each category.

> **Agent rule: Use after minimum competence has been established in each component skill. Avoid presenting interleaving as random topic switching.**

Worked examples and example-problem pairs

Evidence: Moderate to strong for novices

Purpose: Reduce unnecessary search when learning a new procedure by studying a complete solution before independent practice.

Operational protocol

1. Study one complete, correct example.
2. Explain each step and its purpose.
3. Solve a similar problem with partial guidance.
4. Fade guidance and move to independent problems.

Good fit: New mathematical procedures, algorithms, programming, physics, accounting, and other structured problem-solving domains.

Limits and failure modes: Copying examples passively is weak. Guidance should fade as expertise increases.

> **Agent rule: Prefer for low prior knowledge or repeated failure. Transition to retrieval and independent problem solving as performance improves.**

Self-explanation

Evidence: Moderate

Purpose: Generate explanations that connect steps, principles, prior knowledge, and reasons.

Operational protocol

1. Pause after a paragraph, example, or solution step.
2. Explain what it means in your own words.
3. State why the step is valid and how it connects to the objective.
4. Identify uncertainty explicitly and verify it.

Good fit: Conceptual learning, worked examples, proofs, code comprehension, and debugging reasoning.

Limits and failure modes: Unverified explanations may contain misconceptions. Feedback or source checking is necessary.

> **Agent rule: Prompt for short explanations and ask the learner to mark confidence. Do not grade eloquence as mastery.**

Pomodoro / structured focus-break cycles

Evidence: Limited to moderate as a time-management format

Purpose: Support task initiation, sustained attention, and fatigue management by alternating bounded work periods and breaks.

Operational protocol

1. Choose a concrete task before starting.
2. Select a focus interval, commonly 25 minutes, but allow adaptation.
3. Work on one task and record interruptions.
4. Take a short break away from the task.
5. After several cycles, take a longer break.

Good fit: Task initiation, distraction management, routine building, and learners who benefit from visible time boundaries.

Limits and failure modes: The classic 25/5 ratio is not universally optimal. Timers can interrupt deep work, laboratory work, long proofs, or flow states. Pomodoro does not replace an effective cognitive strategy.

> **Agent rule: Treat duration as configurable. Pair the interval with a cognitive method such as retrieval practice, self-explanation, or problem solving.**

3. Recommended product workflow

The product should replace pre-generated rigid study sessions with a user-initiated study flow. The agent can prioritize what should be studied, but the learner chooses when to begin and may choose or accept a suggested method.

Stage	Product behavior	Data produced
1. Select study target	Agent ranks discipline, assessment, and content using deadlines, assessment weight, mastery gap, risk, and available ...	discipline_id, content_id, assessment_id, risk, ...
2. Choose method	Show 2-3 suitable methods with concise rationale. User selects one or accepts the default.	method_id, suggested_by_agent, accepted_suggestion
3. Configure format	Optional timer and break structure. Pomodoro is one format, not the learning activity itself.	planned_focus_minutes, planned_break_minutes
4. Study	Run timer if selected; allow pause, interruption logging, and early completion.	actual_minutes, pauses, interruptions, completion_status
5. Evaluate	Collect subjective and objective outcomes using a short post-session form.	ratings, self-test score, perceived difficulty, notes
6. Adapt	Update method recommendations only after enough comparable observations.	method-context statistics, confidence, next recommendation

4. Post-session evaluation schema

Keep the form brief enough to preserve adherence. Objective measures are preferable when available. The fields below are suitable for structured storage and later analysis.

Field	Type / range	Interpretation
focus_rating	1-5	How consistently the learner stayed on task.
fatigue_rating	1-5	Mental fatigue at the end of the session.
perceived_effectiveness	1-5	Learner's immediate judgment; useful but not proof of learning.
method_fit	1-5	How suitable the method felt for this task.
completion_ratio	0.0-1.0	Completed work divided by planned work.
self_test_score	0.0-1.0 or null	Immediate objective check when a valid self-test exists.
delayed_recall_score	0.0-1.0 or null	Later check; more informative for retention.
interruptions	integer >= 0	Count of external or self-initiated interruptions.
actual_minutes	integer > 0	Observed duration, not only planned duration.
task_type	enum	factual, conceptual, problem_solving, programming, reading, writing, project.
difficulty_before / after	1-5	Perceived task difficulty before and after.
free_text_note	optional text	Short qualitative context; do not require it every time.

5. Adaptation rules for the agent

Do not declare a personal best method after one session. Require at least three comparable sessions per method and task type before forming a low-confidence preference.

Maintain separate scores. Store adherence/preference, immediate performance, and delayed retention separately.

Use context-specific recommendations. A method may be preferred for reading but inferior for programming problems.

Prioritize academic need before method preference. Deadline, assessment weight, mastery gap, prerequisites, and failure risk determine what to study; method history influences how to study it.

Expose uncertainty. Use labels such as insufficient data, tentative, or supported by repeated observations.

Avoid learning-style claims. Do not infer that the learner is a visual, auditory, or kinesthetic learner.

Preserve human control. The learner may override the suggested method and timer duration; the override becomes data, not an error.

Suggested method score

method_score(context) = 0.30 x delayed_performance + 0.25 x immediate_performance + 0.20 x completion + 0.15 x focus + 0.10 x method_fit. Use only available normalized fields and renormalize weights when a field is missing. This is a product heuristic, not a validated psychometric scale.

6. Retrieval-ready knowledge records

ID: retrieval_practice
Knowledge: Use after initial exposure. Ask the learner to recall, solve, or explain without notes; then provide feedback. Strong evidence for long-term retention.

ID: distributed_practice
Knowledge: Schedule repeated encounters over time. Combine with retrieval. Avoid massing all study immediately before the exam.

ID: interleaving
Knowledge: Mix related problem types when the learner already knows the basics. Useful for choosing among strategies; can feel harder initially.

ID: worked_examples
Knowledge: Use for novices and new procedures. Explain steps, then fade guidance toward independent practice.

ID: self_explanation
Knowledge: Ask why a step is valid and how it connects to prior knowledge. Verify explanations to prevent misconception.

ID: pomodoro
Knowledge: Use as optional focus-break structure. Default 25/5 may be adjusted. Pair with a cognitive method; do not claim Pomodoro alone improves retention.

7. Evidence base and references

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Evidence interpretation note

The strongest evidence in this catalog concerns retrieval practice and distributed practice. Interleaving, worked examples, and self-explanation are valuable when matched to task and prior knowledge. Pomodoro has a narrower evidence base and should be represented as an optional attention and time-management structure, not as a universal learning strategy.